

Skills Summary

Rewriting Circles and Parabolas in Standard Form

Use this reference while completing your worksheet or when studying.

Skill 1 — Completing the square on a parabola

The move: to complete $x^2 + px$, add $(p/2)^2$. Since you may not change the equation, subtract the same amount straight back: $x^2 + px + c = (x + p/2)^2 - (p/2)^2 + c$.

Target: $y = a(x - h)^2 + k$, whose vertex is (h, k) — the whole reason for the rewrite. Note the sign: $(x + 3)^2$ means $h = -3$.

Example: Rewrite $y = x^2 + 8x + 10$ in vertex form.

Step 1. Vertex form needs a perfect square, and the only thing standing in the way is the constant — so build the square the x terms want and repay it immediately.

Step 2. Half of 8 is 4, so add and subtract 16: $y = (x^2 + 8x + 16) - 16 + 10$.

$$\Rightarrow y = (x + 4)^2 - 6 \quad (\text{vertex } (-4, -6))$$

Try it: Rewrite $y = x^2 - 6x + 2$ in vertex form.

$$y - 2 = (x - 3)^2 - 7 \quad \text{check:}$$

Skill 2 — Completing the square on a circle

Two squares, one equation. Group the x terms and the y terms, complete each square separately, and add *both* completing constants to the other side.

Target: $(x - h)^2 + (y - k)^2 = r^2$, centre (h, k) , radius r . The right-hand side is r^2 , so square-root it before reporting a radius.

Example: Rewrite $x^2 + y^2 - 2x + 6y - 6 = 0$ in standard form.

Step 1. Neither variable is in a square yet, so complete both — and move the loose constant across first so the bookkeeping stays visible.

Step 2. Half of -2 is -1 (add 1); half of 6 is 3 (add 9): $(x^2 - 2x + 1) + (y^2 + 6y + 9) = 6 + 1 + 9$.

$$\Rightarrow (x - 1)^2 + (y + 3)^2 = 16 \quad (\text{centre } (1, -3), \text{ radius } 4)$$

Try it: Rewrite $x^2 + y^2 + 4x - 8y + 4 = 0$ in standard form.

$$(x + 2)^2 + (y - 4)^2 = 4 \quad \text{check:}$$

Skill 3 — Harder squares, and the feature you read off

Factor before you complete. For $y = ax^2 + bx + c$, pull a out of the terms containing x only: $y = a(x^2 + \frac{b}{a}x) + c$. Complete the square inside the bracket, then multiply the correction back out by a as it leaves.

For a circle the leading coefficients on x^2 and y^2 are equal, so divide the whole equation by that number first and the problem becomes Skill 2.

Example: Rewrite $y = 2x^2 - 8x + 3$ in vertex form, and give the vertex height.

Step 1. The square cannot be completed while a 2 multiplies x^2 , so factor it out of the x terms first and leave the constant outside untouched.

Step 2. $y = 2(x^2 - 4x) + 3 = 2((x - 2)^2 - 4) + 3 = 2(x - 2)^2 - 8 + 3$

Step 3. Vertex height: at $x = 2$, $y = 2(4) - 8(2) + 3 = 8 - 16 + 3$.

$$\Rightarrow y = 2(x - 2)^2 - 5, \quad \text{vertex height } -5$$

Try it: Rewrite $y = 3x^2 - 6x + 4$ in vertex form and give the vertex height.

check: $1 + 2(1 - x)^2 = 3$

Skill 4 — Building the equation from features

Standard form is a template. Given centre (h, k) and radius r , write $(x-h)^2 + (y-k)^2 = r^2$ — watch the sign flip, a centre of -3 gives $(x + 3)$.

General form comes from multiplying out and collecting: $x^2 + y^2 + Dx + Ey + F = 0$. Expanding is the reverse of completing the square, so it is also the fastest way to check your work.

Example: Write the circle with centre $(2, -3)$ and radius 5 in general form.

Step 1. Fill the template first, then expand — going straight to general form invites a sign slip on the centre.

Step 2. $(x - 2)^2 + (y + 3)^2 = 25$, and expanding gives $x^2 - 4x + 4 + y^2 + 6y + 9 - 25 = 0$.

$$\Rightarrow x^2 + y^2 - 4x + 6y - 12 = 0$$

Try it: Write the circle with centre $(-1, 4)$ and radius 3 in general form.

check: $0 = 8 + 12 - 2x + 2y + 2x$